

INCORPORATION-PLAN

Incorporation Plan — Design Toolkit around OpenDesign

Date: 2026-08-06 · **Companion to:** CATALOG.md (the verified inventory) · **Builds on:** OPENDESIGN-ADOPTION-PLAN.md, design-research.md, design-system/ (already built), submodules/constitution (staged).

Goal: turn the catalog into a concrete plan for a reusable **design toolkit** — nested under the HelixConstitution submodule where it belongs, wired into vasic and reusable across all future projects — that combines with OpenDesign to produce **unique, enterprise-grade, non-repeatable UI/UX across every platform, OS, and app type**.

Decision rule used throughout: does the thing produce *artifacts we vendor* (code/tokens/knowledge that must be version-pinned and license-clean) → **git submodule**; is it a *tool an agent calls at runtime* (no vendored code, works over stdio) → **MCP install**; is it *knowledge an agent should reason with* → **encoded skill** (distilled rules, not copied prose).

0. Guardrails this plan must honor (from governance + existing state)

1. **§11.4.162 — OpenDesign is the token source of truth.** No raw hex/px/rem literals, no hand-authored CSS custom properties or inline styles in site stylesheets. **Consequence for this plan:** the parametric-uniqueness layer (Category 4) does NOT hand-write `var(--*)`. It generates a **DTCG token document** that OpenDesign consumes/emits, OR — where OpenDesign cannot yet emit a construct — it is an **upstream PR to OpenDesign (§11.4.74)**, never a local escape hatch. Style Dictionary is used only as the emitter *inside/behind* OpenDesign's contract, not as a parallel token system.

2. **§11.4.170 / §11.4.168 — every surface proven by host-rendered pixels + PDF validation.** New tools (Chrome DevTools MCP, Playwright MCP, Image-Compare MCP) feed this harness (anti-bluff-web-harness), they don't bypass it.
 3. **Container-only pipelines** (per project-translation-pipeline / HelixTranslate). The token-generator and any image-gen run in a container, keys from `~/api_keys.sh`, never in `repo/config`.
 4. **Reduced-motion / GPU-only / cross-browser progressive-enhancement** (from `design-research.md`) remain hard constraints on anything the design agents emit.
 5. **License hygiene (§ anti-bluff):** only vendor MIT/Apache/BSD. GPL tools stay as *external MCP processes* (not vendored source). No-license items are **not** incorporated until relicensed/cleared.
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1. Proposed repository structure

A single toolkit repo, nested as a submodule of HelixConstitution (mirrors the existing “decouple then nest under constitution” plan in `project-vasic-monorepo`), so every Helix project inherits it:

```

HelixConstitution/
├── design-toolkit/                    # NEW submodule (this
plan's deliverable)                  # Claude skills
│   ├── README.md
│   └── skills/
(vendored, Apache/MIT only)
│   └── _vendor/                      # nested submodules of
upstream skill repos (pinned)
│   │   ├── anthropics-skills/        # submodule → anthropics/
skills (Apache-2.0)
│   │   ├── designer-skills/         # submodule → Owl-
Listener/designer-skills (MIT)
│   │   └── ui-ux-pro-max/            # submodule →
nextlevelbuilder/ui-ux-pro-max-skill (MIT)
│   │   └── jezweb-skills/           # submodule → jezweb/
claude-skills (MIT)
│   │   └── anydesign/                # submodule → uxKero/
anydesign (MIT)
│   │   └── recipes/                 # OUR design-specialist
agent recipes (§4) – the value-add
│       ├── theming-designer/
│       ├── layout-architect/
│       ├── animation-designer/
│       ├── iconographer/
│       ├── ux-flow-designer/
│       └── design-qa-auditor/
└── knowledge/                        # distilled, vendorable
references (§3 encode) – rules not prose

```

```

    |   |— ally/           (WCAG 2.2 gate + WAI-ARIA APG patterns)
    |   |— platforms/     (HIG rules-only checklist, Material 3,
Fluent 2, GNOME)
    |   |— color/         (OKLCH model, WCAG contrast gate, Radix/
Open Color scales)
    |   |— type/          (Utopia fluid formulas, variable-font
axes, MD3 scale roles)
    |   |— layout/        (Every Layout primitives, container-
query method, 8pt grid)
    |   |— motion/        (MD3 motion tokens, easings.net table,
reduced-motion gate)
    |   |— generators/    # the parametric
uniqueness layer (§5)
    |   |— package.json   # deps: material-color-
utilities, colorjs.io, chroma-js,
    |   |                 #         utopia-core,
style-dictionary, @svgsdotjs/svg.js
    |   |— design-dna.mjs  # seed → parameter vector
    |   |— gen-tokens.mjs  # vector → DTCG token doc
(→ OpenDesign consumes/emits)
    |   |— gen-marks.mjs   # seed → build-time
static SVG marks/backgrounds (svg.js)
    |   |— lib/_vendor/    # OPTIONAL submodules if
we want source-pinned instead of npm
    |   |— mcp/
    |   |— mcp.design.json # ready-to-merge MCP
server definitions (§2)
    |   |— docs/
    |       — DESIGN-TOOLKIT.md # how OpenDesign +
generators + skills + MCP combine

```

Then, exactly as the existing decoupled-libs plan does: HelixConstitution is a submodule of vasic, so vasic gets design-toolkit/ transitively. Sites consume only the **emitted OpenDesign tokens** — never reach into generators/ at runtime.

2. What becomes an MCP install (runtime tools — NOT vendored)

Add via `claude mcp add` (project scope for vasic, or a shipped `mcp.design.json` teams merge). None require a paid subscription. GPL/no-license items are kept out entirely; the ones below are all local + permissive.

MCP server	Command	License	Role in the toolkit
	<code>claude mcp add</code>		Design QA:
	<code>chrome-devtools</code>	Apache-2.0	Lighthouse a11y/

MCP server	Command	License	Role in the toolkit
Chrome DevTools MCP	--scope user npx chrome-devtools-mcp@latest		perf, screenshots → feeds §11.4.170 proof (already a plugin here)
Playwright MCP	claude mcp add playwright npx @playwright/mcp@latest	Apache-2.0	Screenshot / visual-regression backbone for the proof harness
shadcn MCP (official)	npx shadcn@latest mcp init --client claude	MIT	Component discovery/reuse before building custom
shadcn-ui-mcp-server (Jpisnice)	claude mcp add shadcn -- npx -y @jpisnice/shadcn-ui-mcp-server	MIT	Multi-framework (React/Vue/Svelte/RN) component source
Framelink Figma MCP	claude mcp add framelink-figma -- npx -y figma-developer-mcp --figma-api-key=\$FIGMA_PAT --stdio	MIT	Design <i>ingestion</i> from Figma (free PAT) when a project starts from Figma
draw.io MCP (lgazo)	npx -y drawio-mcp-server --editor	MIT	Architecture/diagram authoring (case-study spine in design-research.md)
Mermaid MCP (peng-shawn)	claude mcp add mermaid -- npx -y @peng-shawn/mermaid-mcp-server	MIT	Agent Mermaid → rendered PNG/SVG diagram assets
google-fonts-mcp	claude mcp add google-fonts --uvx google-fonts-mcp	MIT	Font pairing → CSS/Tailwind type systems (feeds theming-designer)
sharp-mcp	npx -y sharp-mcp	MIT	Local AVIF/WebP/resize (the responsive-image work in design-research.md §5)

Pilot-then-standardize (low stars — evaluate before committing): x51xxx/coolors-mcp (MD3/OKLCH color ops), kenneives/design-token-bridge-mcp (DTCG→Material3/SwiftUI/CSS), leky90/mcp-image-compare-server (visual diff — but Playwright's toHaveScreenshot is the safer default).

Explicitly NOT installed: Official Figma Dev Mode MCP (paid seat), 21st.dev Magic MCP (paid/keyed), LottieFiles Creator MCP (hosted account). Iconify MCPs are GPL/no-license → prefer the **iconographer skill + local Iconify data** over an MCP (see §4).

3. What becomes a vendored submodule (pinned source we ship)

3a. Skill repos (nested submodules under design-toolkit/skills/_vendor/, pinned to a reviewed SHA): - anthropics/skills (Apache-2.0) — take frontend-design, theme-factory, web-artifacts-builder, brand-guidelines, canvas-design, algorithmic-art.

Foundation. - Owl-Listener/designer-skills (MIT) — vendor the *subset* that complements OpenDesign (design-research, ux-strategy, visual-critique); skip anything duplicating our recipes. - nextlevelbuilder/ui-ux-pro-max-skill (MIT) — for cross-platform stack coverage (Compose/Flutter/SwiftUI/WinUI/Avalonia). Use as a *reference the recipes cite*, gated by OpenDesign tokens so it doesn't drift generic. - jezweb/claude-skills (MIT) — frontend + design-assets (palette/favicon/icon/image). - uxKero/anydesign (MIT) — design→token extraction (ingestion counterpart to OpenDesign).

3b. Generator libraries (npm deps by default; optionally source-pinned submodules under generators/lib/_vendor/): material-color-utilities (Apache-2.0), colorjs.io (MIT), chroma-js (BSD-3), style-dictionary (Apache-2.0), @svgsdotjs/svg.js (MIT), utopia-core (**confirm LICENSE first** — sibling is MIT). Optional: @adobe/leonardo-contrast-colors (Apache-2.0) for contrast-driven theme modes.

3c. Knowledge (design-toolkit/knowledge/ — *distilled rules we author*, citing sources; we do NOT copy copyrighted prose): encode WCAG 2.2 + WAI-ARIA APG (open), Material 3 (open, richest), DTCCG spec (open), easings.net table (MIT), Radix/Open Color scales (MIT); paraphrase Apple HIG / Fluent 2 / Atomic Design / Every Layout primitives into checklists. Each file: source URL, license note, and the machine-usable rules.

4. Design-specialist agent recipes (the toolkit's core value-add)

Each recipe = a skill (SKILL.md + supporting refs) defining what the specialist **knows** (which vendored knowledge/skills it draws on), **the tools it uses** (which MCP servers), **what it does** (procedure), and **how it respects the guardrails**. All emit through OpenDesign; none hand-author tokens.

4.1 theming-designer — the uniqueness engine

- **Knows:** color science (knowledge/color/ — OKLCH model, WCAG contrast gate, Radix/Open Color), Material 3 color roles, the parametric method (Category 4), font-pairing→personality matrix, DTCG.
- **Uses:** generators/ (material-color-utilities + colorjs.io/chroma-js + utopia-core → Style Dictionary → OpenDesign), google-fonts-mcp, colors-mcp (pilot).
- **Does:** takes a project **design-DNA seed+ brand adjectives** → picks {seedHue, mcuVariant, typeRatio, spaceMultiplier, radiusBase, fontPairId, contrastMode} from bounded ranges (constrained selection, justified — per the CPT research finding) → generates a full accessible token set → hands DTCG to OpenDesign to emit → verifies WCAG contrast on every semantic pair. **This is what makes each project distinct-but-on-brand and reproducible.**
- **Guardrail:** never writes var(--*) into site CSS; if OpenDesign can't emit a needed construct → upstream PR.

4.2 layout-architect

- **Knows:** Every Layout primitives (Stack/Cluster/Sidebar/Switcher/Cover/Grid — concepts), container-query-first responsiveness, 8pt grid, Apple HIG platform-adaptation rules (safe areas, ≥44pt targets, native nav), Material 3 layout regions.
- **Uses:** shadcn MCPs (component reuse), Chrome DevTools MCP (responsive checks at real viewports).
- **Does:** designs page/screen structure from composable primitives + container queries (not viewport-only breakpoints); applies per-platform adaptation checklist for cross-platform nativeness; expresses breakpoints/regions as OpenDesign layout tokens.
- **Guardrail:** progressive-enhancement base must render without JS; verify by rendering+clicking at phone/tablet/desktop (the lesson already learned in project-design-state).

4.3 animation-designer

- **Knows:** MD3 motion tokens, easings.net table, Disney principles (CC-BY-SA summary), the GPU-only / reduced-motion / deterministic-end-state rules from design-research.md.
- **Uses:** Chrome DevTools MCP (perf trace / smoothness), Playwright MCP (force deterministic end-state for golden capture).
- **Does:** specifies micro-interactions and scroll/view-transition motion as tokenized CSS/SVG (drop Lottie for simple motion); animates only transform/opacity/compositor-safe filter; every motion gated on prefers-reduced-motion: no-preference; provides the reduced-motion capture variant for §11.4.170.
- **Guardrail:** CI greps emitted CSS to fail on animated width/height/top/left/box-shadow.

4.4 iconographer

- **Knows:** icon-system theory (stroke/weight/grid consistency, optical sizing), the self-hosted sprite approach already in design-system/icons/icons.svg.
- **Uses:** local **Iconify data** + jezebel design-assets skill for SVG icon sets (prefer this over the GPL/no-license icon MCPs); sharp-mcp/SVGOM for optimization.
- **Does:** selects/uses a consistent icon set per project personality, generates a self-hosted subset sprite (no remote requests, no CLS), colors from tokens, ensures a11y labels.
- **Guardrail:** no remote icon CDNs (matches the self-hosted-fonts/no-Google-Fonts discipline already enforced).

4.5 ux-flow-designer

- **Knows:** UX-strategy frameworks (from designer-skills + the *concepts* of awesome-ux-skills, once its license is cleared), the case-study spine (Problem→Approach→Architecture→Outcome→Artifact) and trust/proof devices from design-research.md, WAI-ARIA APG interaction models.
- **Uses:** draw.io/Mermaid MCP (flow + architecture diagrams), Framelink Figma MCP (ingest existing flows), anydesign skill (extract structure from references).
- **Does:** designs information architecture, navigation, and page/flow templates; produces the repeatable portfolio/product spine; specifies keyboard-interaction patterns per APG.
- **Guardrail:** every interactive element gets a real behavior + a11y semantics (the “behavior not href” lesson from interactive-behavior.spec.js).

4.6 design-qa-auditor

- **Knows:** WCAG 2.2 success criteria as a hard gate, §11.4.170/.168 proof requirements, CWV thresholds.
- **Uses:** Chrome DevTools MCP (Lighthouse a11y/perf), Playwright MCP + Image-Compare (visual regression), the existing anti-bluff-web-harness.
- **Does:** audits each screen × state × {light,dark} by rendered pixels + a11y + contrast + CWV; fails the build on regressions; produces the evidence bundle.
- **Guardrail:** verify by rendering+clicking at real viewports, not structural/heuristic checks (institutionalized lesson).

Shared substrate: all recipes read design-toolkit/knowledge/ and defer token authority to OpenDesign. anthropics/skills:frontend-design + theme-factory are the aesthetic/theming base the recipes build on; ui-ux-pro-max supplies cross-platform stack specifics on demand.

5. The parametric uniqueness layer + OpenDesign — how they combine

Pipeline (all container-only, keys from ~/api_keys.sh):

```
project seed (= hash(project-name+domain) or chosen brand hue)
  | design-dna.mjs
  ▼
parameter vector {seedHue, mcuVariant, typeRatio, spaceMultiplier,
radiusBase, fontPairId, contrastMode}
  | gen-tokens.mjs
  |─ material-color-utilities → HCT tonal palettes + role-
mapped light/dark scheme (variant = personality)
  |─ colorjs.io / chroma-js → custom OKLCH harmony ramps,
gamut-safe P3 (+ sRGB fallback)
  |─ utopia-core → fluid type/space clamp()
scales from ratio + multiplier
  └─ radius + density tokens
  ▼
  DTCG token document → OpenDesign (consumes/emits) →
tokens.css / per-platform output
  |
  ($11.4.162 source of truth;
Style Dictionary sits behind it)
  ▼
  sites/PDFs consume ONLY the emitted OpenDesign tokens
+
gen-marks.mjs (svg.js, seeded RNG, BUILD TIME) → static per-
project SVG marks/backgrounds/dividers (colors from tokens)
```


Why this yields uniqueness without chaos: OpenDesign owns *structure and semantics* (component set, token names, light/dark, a11y); the generator supplies *values* from a different point in the parameter space per project; the seeded SVG layer adds a *visual signature*. Two Vasic projects can share the exact component library and still read as unmistakably different brands — and re-running the same seed reproduces the same design exactly (deterministic). This is the Material-3-dynamic-color pattern (proven, Apache-2.0) generalized to type/space/shape/marks.

§11.4.162 reconciliation (critical): the generator's output is a **DTCG document fed into OpenDesign**, not hand-written CSS. If OpenDesign cannot yet ingest generator-produced tokens or emit a construct (e.g. `light-dark()`, a specific `clamp()` shape), that gap is an **upstream OpenDesign PR (§11.4.74)** — the generator never writes `var(--*)` or literals into a site stylesheet. Confirm during Phase A that OpenDesign's token-ingestion contract accepts an external DTCG doc; if not, that's the first upstream PR.

6. Sequenced rollout (small, verifiable increments)

- **Phase A — Scaffold + prove the pipeline (vertical slice).** Create design-toolkit/ skeleton; add the 2 highest-value MCP installs (Chrome DevTools already present; add Playwright, shadcn). Build generators/ with material-color-utilities + colorjs.io + utopia-core + Style Dictionary; produce ONE token set from a seed and prove OpenDesign can emit it (or file the upstream PR). Confirm utopia-core LICENSE before pinning. Verify with §11.4.170 render of the existing button slice under a generated theme.
- **Phase B — Vendor the skill foundation.** Nest anthropics/ skills + Owl-Listener/designer-skills as pinned submodules; author the theming-designer and design-qa-auditor recipes (the two with immediate leverage). Wire google-fonts-mcp, sharp-mcp.
- **Phase C — Cross-platform + remaining recipes.** Add ui-ux-pro-max + jezweb + anydesign submodules; author layout-architect, animation-designer, iconographer, ux-flow-designer. Add draw.io/Mermaid/Framelink MCPs. Encode the knowledge/reference set.
- **Phase D — Generative marks + distinctness proof.** Add gen-marks.mjs (svg.js) build-time signature layer; generate a *distinct* theme+mark set for milosvasic.ru vs vasic.digital from different seeds and prove via §11.4.170 that they render as different brands on the shared component library.
- **Phase E — Nest under HelixConstitution + wire into vasic.** Move design-toolkit/ under the constitution submodule,

wire back into vasic (mirrors the held “nest decoupled libs under constitution” step); gate CI on design-qa-auditor.

Blocked/held dependencies (from existing state): nesting under HelixConstitution is entangled with the un-pushed §11.4.236 amendment + umbrella push (held for user GO in project-design-state/project-vasic-monorepo). Phases A-D can proceed in-tree under _analysis/design-system without that GO; Phase E waits on it.

7. Honesty summary (what NOT to rely on)

- **No single OSS “brand-in-a-box”** does color+type+space+shape+marks end-to-end — §5 assembly is our work (well-scoped, all deps MIT/Apache/BSD).
- **Low-star MCPs** (coolors, design-token-bridge, image-compare) must be **piloted** before standardizing; don’t gate CI on them until proven.
- **License blockers to clear before vendoring:** utopia-core (confirm), awesome-ux-skills/awesome-list indexes/mcp-universal-icons/several color+icon MCPs (no license — do not vendor). Iconify MCPs are GPL → keep external or use local Iconify data via the iconographer skill instead.
- **Paid/excluded, do not add:** Official Figma Dev Mode MCP, 21st.dev Magic MCP, LottieFiles Creator MCP, Refactoring UI, Every Layout (buy for reference; encode concepts only), APCA as an authoritative gate (advisory only — enforce WCAG 2.2 ratios).
- **Copyrighted knowledge** (Apple HIG, Fluent 2, Atomic Design): encode paraphrased rules, never vendor the prose.